

# Nonlinear Dynamics and Chaos with Applications to Physics, Biology, Chemistry, and Engineering 2nd Edition

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## 2 Flows on the Line

### 2.1 A Geometric Way of Thinking

#### 2.1.1

For a fixed point of the flow  $\dot{x} = \sin x$  on the line, we have

$$\dot{x} = 0 \Rightarrow \sin x = 0 \Rightarrow x = n\pi, \quad n \in \mathbb{Z}.$$

Thus all fixed points are given by  $x = n\pi$ ,  $n \in \mathbb{Z}$ .

#### 2.1.2

The points  $x$  for which the flow has the greatest velocity to the right are those for which  $\dot{x} > 0$  and  $x$  is a local maximum. These are given by

$$x = \frac{(4n+1)\pi}{2}, \quad n \in \mathbb{Z}.$$

#### 2.1.3

(a) We have

$$\dot{x} = \sin x,$$

so that

$$\begin{aligned}
 \ddot{x} &= \frac{d}{dt}(\sin x) \\
 &= \dot{x} \cos x \\
 &= \sin x \cos x \\
 &= \frac{1}{2} \sin(2x).
 \end{aligned}$$

- (b) The maximum positive acceleration is given by the local maxima of  $\ddot{x} = (1/2) \sin(2x)$ , i.e., where

$$2x = \frac{(4n+1)\pi}{2} \Rightarrow x = \frac{(4n+1)\pi}{4}, \quad n \in \mathbb{Z}.$$

#### 2.1.4

- (a) We begin by evaluating

$$t = \ln \left| \frac{\csc x_0 + \cot x_0}{\csc x + \cot x} \right|$$

with  $x_0 = \pi/4$  to find

$$\begin{aligned}
 t &= \ln \left| \frac{\sqrt{2} + 1}{\csc x + \cot x} \right| \\
 &= \ln \left( \frac{1 + \sqrt{2}}{\csc x + \cot x} \right).
 \end{aligned}$$

Then we apply the function  $e^z$  to both sides of to find

$$\begin{aligned}
 e^t &= \frac{1 + \sqrt{2}}{\csc x + \cot x} \\
 &= \frac{1 + \sqrt{2}}{\cot(x/2)} \\
 \Leftrightarrow \cot(x/2) &= \frac{1 + \sqrt{2}}{e^t} \\
 \Leftrightarrow \frac{1}{\tan(x/2)} &= \frac{1 + \sqrt{2}}{e^t} \\
 \Leftrightarrow \tan(x/2) &= \frac{e^t}{1 + \sqrt{2}} \\
 \Leftrightarrow \frac{x}{2} &= \tan^{-1} \left( \frac{e^t}{1 + \sqrt{2}} \right) \\
 \Leftrightarrow x &= 2 \tan^{-1} \left( \frac{e^t}{1 + \sqrt{2}} \right).
 \end{aligned}$$

Now, we have

$$\lim_{t \rightarrow \infty} x(t) = \lim_{t \rightarrow \infty} \left[ 2 \tan^{-1} \left( \frac{e^t}{1 + \sqrt{2}} \right) \right] = \pi,$$

since  $\lim_{t \rightarrow \infty} e^t = \infty$  and  $\lim_{t \rightarrow \infty} \tan^{-1} t = \pi$ .

- (b) An analytic solution is given by performing the steps in (a) without assuming  $x_0 = \pi/4$ , so that we have

$$\begin{aligned} x(t) &= 2 \tan^{-1} \left( \frac{e^t}{\csc x_0 + \cot x_0} \right) \\ &= 2 \tan^{-1} \left( \frac{e^t}{\cot(x_0/2)} \right). \end{aligned}$$

### 2.1.5

- (a) Since the time derivative oscillates with period  $2\pi$ , a mechanical system which is approximately governed by  $\dot{x} = \sin x$  is a pendulum, with angle of relative to the horizontal axis given by  $x$  and  $x = \pi$  the lowest point in the pendulum's trajectory, i.e., an upside-down Cartesian coordinate system.
- (b) Since it will always fall back towards the vertical axis when it is suspended perpendicular to the horizontal axis,  $x^* = 0$  is an unstable fixed point, while  $x^* = \pi$  corresponds to the lowest point in the trajectory, so that the pendulum will not move if it starts at this point.

## 2.2 Fixed Points and Stability

### 2.2.1

In Figure 2.1, we have a plot of the vector field, fixed points, and various graphs with different initial conditions for the equation

$$\dot{x} = 4x^2 - 16$$

An explicit solution for  $\dot{x} = 4x^2 - 16$  can be found by re-writing the equation as

$$\frac{dx}{dt} = 4x^2 - 16,$$

so that we have separable equations. Then, we may write

$$\frac{dx}{4x^2 - 16} = dt,$$

so that by using partial fraction decomposition, we have

$$\left[ \frac{\frac{1}{16}}{x-2} - \frac{\frac{1}{16}}{x+2} \right] dx = dt,$$

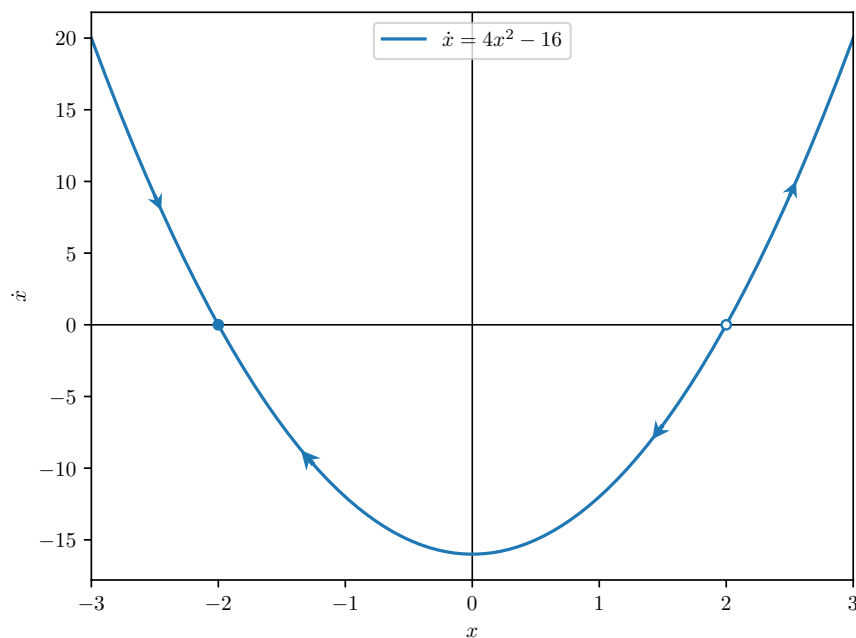


Figure 2.1: Vector field and fixed points for  $\dot{x} = 4x^2 - 16$

and integrate either sides with respect to the  $x$  and  $t$ , respectively, so that we have

$$\begin{aligned}
 & \int \left[ \frac{\frac{1}{16}}{x-2} - \frac{\frac{1}{16}}{x+2} \right] dx = \int dt \\
 \Rightarrow & \frac{1}{16} \ln|x-2| - \frac{1}{16} \ln|x+2| = t + c_1 \\
 \Rightarrow & \ln \left| \frac{x-2}{x+2} \right| = 16t + c_2 \\
 \Rightarrow & \frac{x-2}{x+2} = \pm e^{16t+c_2}.
 \end{aligned}$$

After replacing  $\pm e^{c_2}$  by  $c$  and solving the last equation for  $x$ , we have the one-parameter family of solutions

$$x = 2 \frac{1 + ce^{16t}}{1 - ce^{16t}}.$$

If we let  $x(0) = 0$ , we have

$$\begin{aligned} 2\frac{1+c}{1-c} &= 0 \\ \Rightarrow 1+c &= 0 \\ \Rightarrow c &= -1, \end{aligned}$$

so that

$$x_1(t) = 2\frac{1-e^{16t}}{1+e^{16t}}$$

is a solution corresponding to the initial condition  $x(0) = 0$ .

Likewise, if we let  $x(0) = 4$ , we have

$$\begin{aligned} 2\frac{1+c}{1-c} &= 4 \\ \Rightarrow 1+c &= 2(1-c) \\ \Rightarrow c &= \frac{1}{3}, \end{aligned}$$

so that

$$x_2(t) = 2\frac{1+\frac{1}{3}e^{16t}}{1-\frac{1}{3}e^{16t}}$$

is a solution corresponding to the initial condition  $x(0) = 4$ .

Finally, if we let  $x(0) = -4$ , we have

$$\begin{aligned} 2\frac{1+c}{1-c} &= -4 \\ \Rightarrow 1+c &= -2(1-c) \\ \Rightarrow c &= 3, \end{aligned}$$

so that

$$x_3(t) = 2\frac{1+3e^{16t}}{1-3e^{16t}}$$

is a solution corresponding to the initial condition  $x(0) = -4$ .

In Figure 2.2, we have a plot of these solutions for the given initial conditions.

### 2.2.2

In Figure 2.3, we have a plot of the vector field, fixed points, and various graphs with different initial conditions for the equation

$$\dot{x} = 1 - x^{14}$$

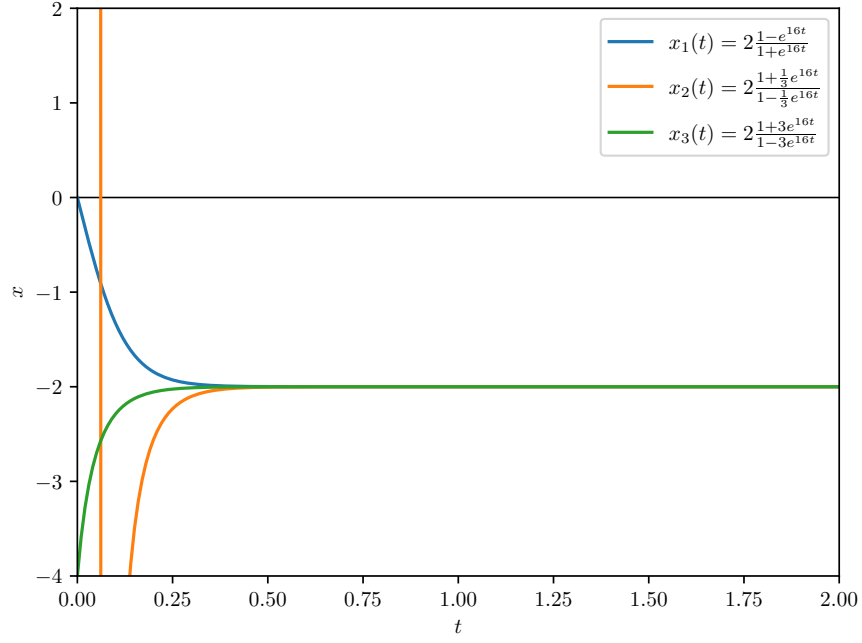


Figure 2.2: Graph of solutions of  $\dot{x} = 4x^2 - 16$

Again, we separate the equation to find

$$\frac{dx}{1 - x^{14}} = dt.$$

Integrating either side with respect to  $x$  and  $t$ , respectively, we have

$$\begin{aligned} \int \frac{dx}{1 - x^{14}} &= \int dt \\ \Rightarrow \ln |1 - x^{14}| &= t + c_1 \\ \Rightarrow 1 - x^{14} &= \pm e^{t+c_1} \\ \Rightarrow x^{14} &= \pm e^{t+c_1} + 1 \\ \Rightarrow x^{14} &= ce^t + 1 \\ \Rightarrow x &= \pm \sqrt[14]{ce^t + 1}, \end{aligned}$$

for  $ce^t + 1 \geq 0$ .

It does not seem possible to find a simple closed-form solution to this ODE.

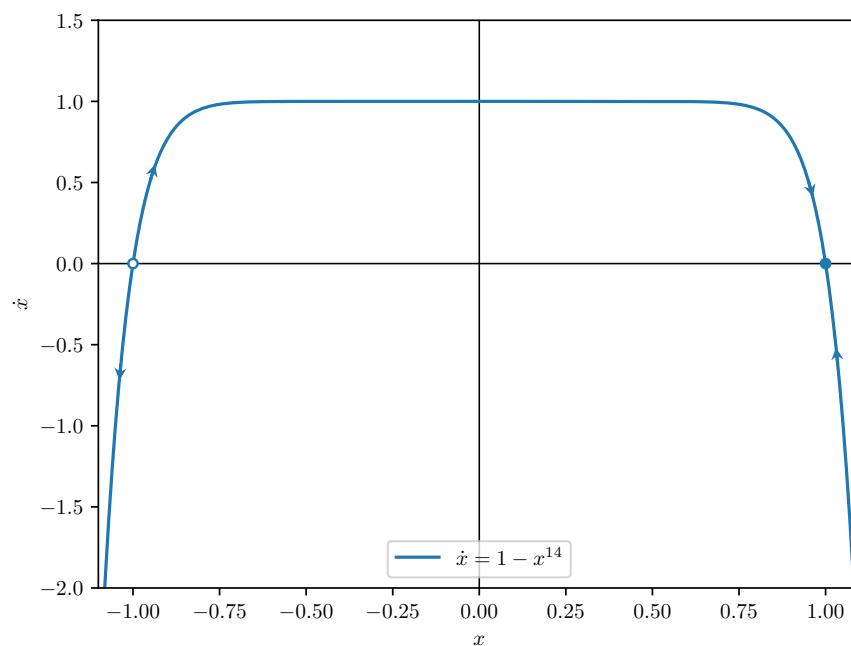


Figure 2.3: Vector field and fixed points for  $\dot{x} = 1 - x^{14}$

### 2.2.3

In Figure 2.4, we have a plot of the vector field, fixed points, and various graphs with different initial conditions for the equation

$$\dot{x} = x - x^3$$

We use the method of separable equations again to find

$$\frac{dx}{x - x^3} = dt,$$

and using partial fraction decomposition, we have

$$\left[ \frac{1}{x} + \frac{\frac{1}{2}}{1-x} - \frac{\frac{1}{2}}{1+x} \right] dx = dt.$$

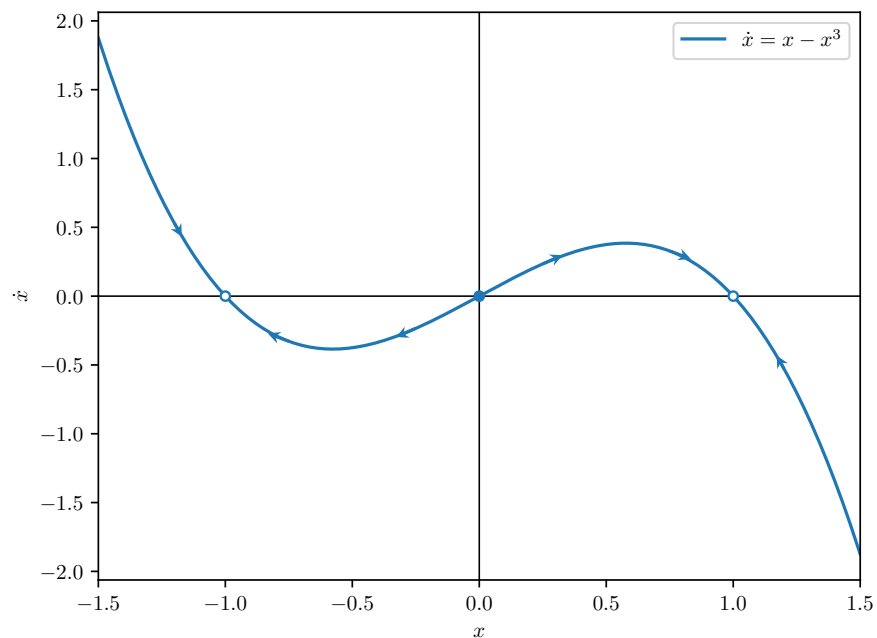


Figure 2.4: Vector field and fixed points for  $\dot{x} = x - x^3$

Integrating both sides yields

$$\begin{aligned}
 \int \frac{dx}{x} + \frac{1}{2} \int \frac{dx}{1-x} - \frac{1}{2} \int \frac{dx}{1+x} &= \int dt \\
 \Leftrightarrow \ln |x| + \frac{1}{2} \ln |1-x| - \frac{1}{2} \ln |1+x| &= t + c_1 \\
 \Leftrightarrow \ln \left| \frac{x^2(1-x)}{1+x} \right| &= 2t + c_2 \\
 \Leftrightarrow \frac{x^2(1-x)}{1+x} &= \pm e^{2t+c_2} \\
 &= ce^{2t}.
 \end{aligned}$$

Again, it does not seem possible to find a simple closed-form solution to this ODE.

### 2.2.4

In Figure 2.5, we have a plot of the vector field, fixed points, and various graphs with different initial conditions for the equation

$$\dot{x} = e^{-x} \sin x$$

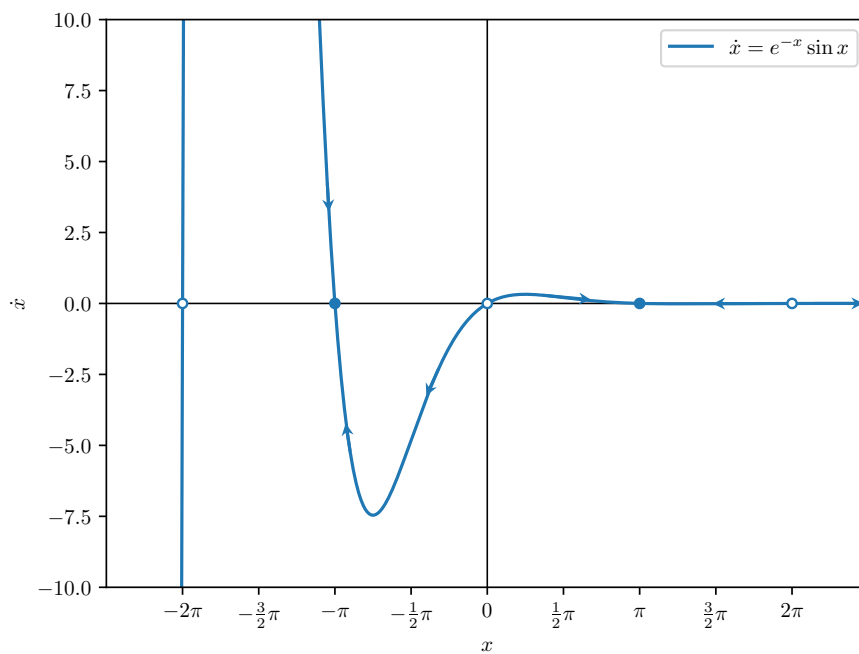


Figure 2.5: Vector field and fixed points for  $\dot{x} = e^{-x} \sin x$

We separate equations to find

$$\int e^x \csc x \, dx = \int dt,$$

but since  $e^x \csc x$  is not an elementary function, we cannot find a simple, closed-form solution.

### 2.2.5

In Figure 2.6, we have a plot of the vector field and various graphs with different initial conditions for the equation

$$\dot{x} = 1 + \frac{1}{2} \cos x. \quad (2.1)$$

Note that there are no fixed points for Equation (2.1), so that these are not included in Figure 2.6.

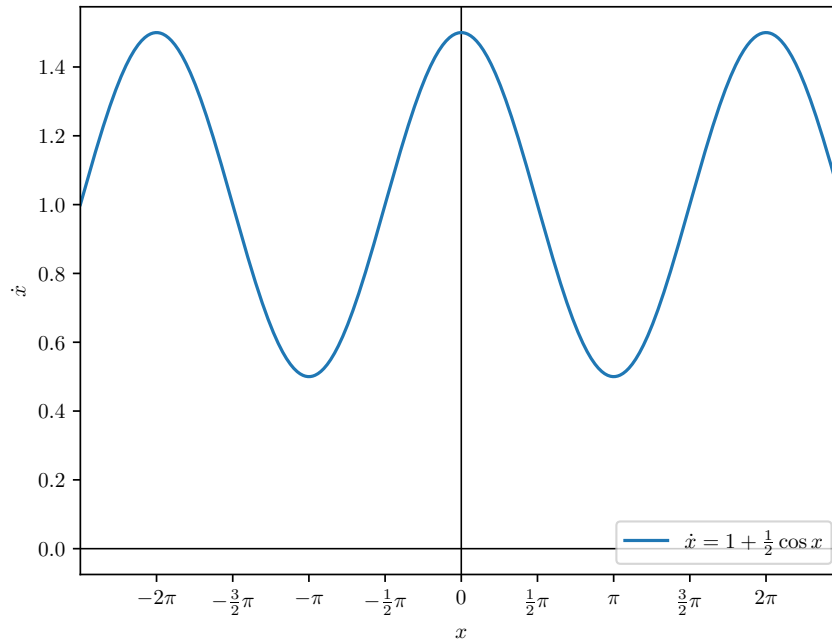


Figure 2.6: Vector field for  $\dot{x} = 1 + \frac{1}{2} \cos x$

We separate equations to find

$$\begin{aligned} \frac{dx}{dt} &= 1 + \frac{1}{2} \cos x \\ \Leftrightarrow \int \frac{dx}{1 + \frac{1}{2} \cos x} &= \int dt. \end{aligned}$$

We introduce a new variable,  $u = \tan \frac{x}{2}$ , so that we have

$$\cos x = \frac{1 - u^2}{1 + u^2},$$

and

$$dx = \frac{2}{1 + u^2} dt.$$

Then we have

$$\begin{aligned}\int \frac{dx}{1 + \frac{1}{2} \cos x} &= 4 \int \frac{du}{3 + u^2} \\ &= \frac{4}{\sqrt{3}} \tan^{-1} \left( \frac{u}{\sqrt{3}} \right) + c_1 \\ &= \frac{4}{\sqrt{3}} \tan^{-1} \left( \frac{\tan \frac{x}{2}}{\sqrt{3}} \right) + c_1.\end{aligned}$$

Thus our ordinary differential equation becomes

$$\frac{4}{\sqrt{3}} \tan^{-1} \left( \frac{\tan \frac{x}{2}}{\sqrt{3}} \right) = t + c_2,$$

so that we have

$$\begin{aligned}\tan^{-1} \left( \frac{\tan \frac{x}{2}}{\sqrt{3}} \right) &= \frac{\sqrt{3}}{4} (t + c_2) \\ \Leftrightarrow \frac{\tan \frac{x}{2}}{\sqrt{3}} &= \tan \left( \frac{\sqrt{3}}{4} t + c_3 \right) \\ \Leftrightarrow \tan \frac{x}{2} &= \sqrt{3} \tan \left( \frac{\sqrt{3}}{4} t + c_3 \right) \\ \Leftrightarrow x &= 2 \tan^{-1} \left[ \sqrt{3} \tan \left( \frac{\sqrt{3}}{4} t + c_3 \right) \right],\end{aligned}$$

where  $c_3 = \frac{\sqrt{3}}{4} c_2$ .

Substituting the initial condition  $x(0) = 0$  yields

$$\begin{aligned}\frac{4}{\sqrt{3}} \tan^{-1} \left( \frac{\tan 0}{\sqrt{3}} \right) &= c_2 \\ \Leftrightarrow \frac{4}{\sqrt{3}} \tan^{-1} 0 &= c_2 \\ \Leftrightarrow 0 &= c_2,\end{aligned}$$

so that  $c_3 = 0$ , and we have a solution corresponding to the initial condition  $x(0) = 0$  given by

$$x(t) = 2 \tan^{-1} \left[ \sqrt{3} \tan \left( \frac{\sqrt{3}}{4} t \right) \right].$$

In Figure 2.7, we have a plot of this solution.

### 2.2.6

In Figure 2.8, we have a plot of the vector field and fixed points for the equation

$$\dot{x} = 1 - 2 \cos x$$

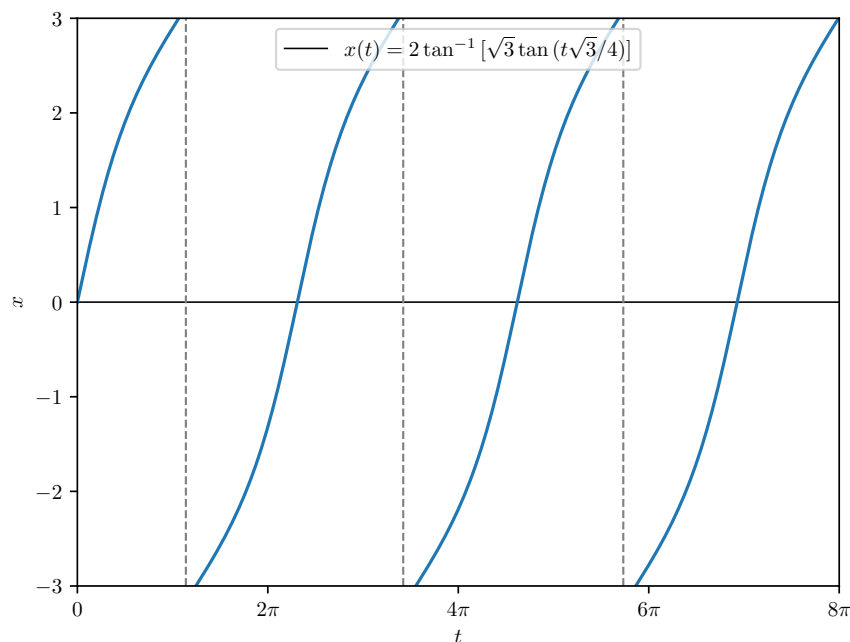


Figure 2.7: Graph of a solution of  $\dot{x} = 1 + \frac{1}{2} \cos x$

### 2.2.7

In Figure 2.9, we have a plot of the functions  $e^x$  and  $\cos x$ . From this figure, we can estimate that the fixed points are given approximately by  $x = -1.25$  and  $x = 0.0$ .

### 2.2.8

We are looking for an equation which is consistent with a phase portrait with a half-stable fixed point at  $x = -1$ , a stable fixed point at  $x = 0$ , and an unstable fixed point at  $x = 2$ . One equation which satisfies these conditions is given by

$$\begin{aligned} f(x) &= x(x+1)^2(x-2) \\ &= x^4 - 3x^2 - 2x. \end{aligned}$$

We plot the vector field and fixed points for this equation in Figure 2.10.

### 2.2.9

From Figure 2, we see that we are looking for an equation  $\dot{x} = f(x)$  with a stable fixed point at  $x = 0$  and an unstable fixed point at  $x = 1$ , whose solutions  $x(t)$

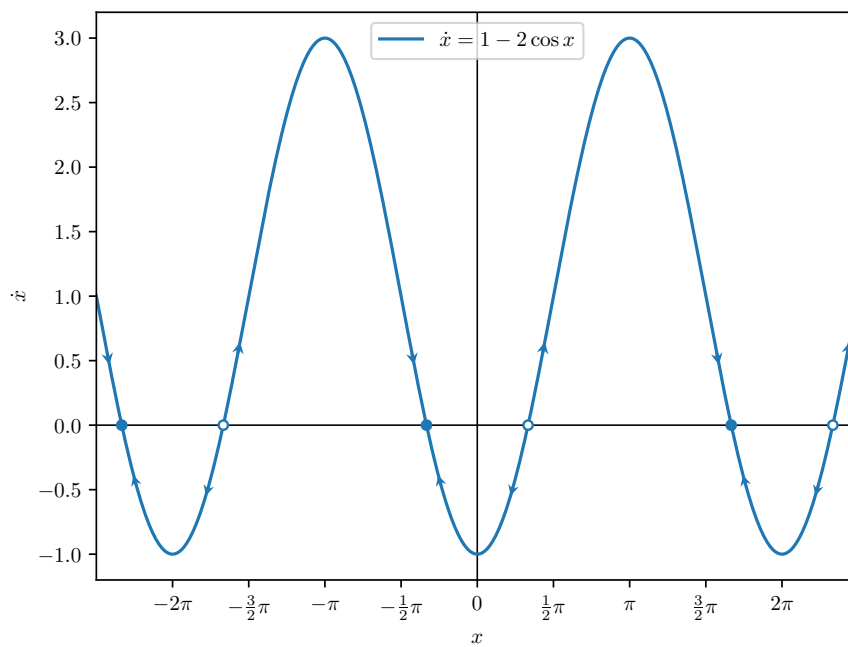


Figure 2.8: Vector field and fixed points for  $\dot{x} = 1 - 2 \cos x$

satisfy the following three conditions:

- (i)  $x_1(t)$  is a particular solution satisfying  $x_1(0) > 1$ , where

$$\lim_{t \rightarrow \infty} x_1(t) = \infty,$$

- (ii)  $x_2(t)$  is a particular solution satisfying  $0 < x_2(0) < 1$ , where

$$\lim_{t \rightarrow \infty} x_2(t) = 0,$$

- (iii)  $x_3(t)$  is a particular solution satisfying  $x_3(0) = -1$ , where

$$\lim_{t \rightarrow \infty} x_3(t) = 0.$$

One such equation is  $\dot{x} = x^2 - x$ , which we solve as follows. First, we

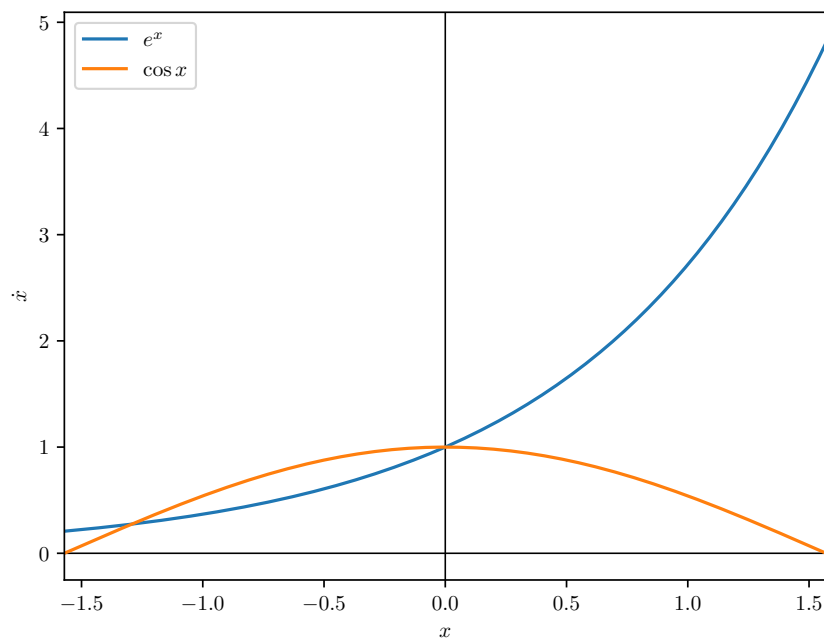


Figure 2.9: Graphs of  $e^x$  and  $\cos x$

integrate both sides with respect to  $x$  and  $t$  to find

$$\begin{aligned}
 \frac{dx}{dt} &= x^2 - x \\
 \Leftrightarrow \int \frac{dx}{x^2 - x} &= \int dt \\
 \Leftrightarrow \int \frac{dx}{x-1} - \int \frac{dx}{x} &= t + c_1 \\
 \Leftrightarrow \ln|x-1| - \ln|x| &= t + c_1 \\
 \Leftrightarrow \ln\left|\frac{x-1}{x}\right| &= t + c_1 \\
 \Leftrightarrow \frac{x-1}{x} &= \pm e^{t+c_1}.
 \end{aligned}$$

If we replace  $\pm e^{c_1}$  by  $c$  and solve for  $y$ , we get the one-parameter family of solutions

$$x = \frac{1}{1 - ce^t}.$$

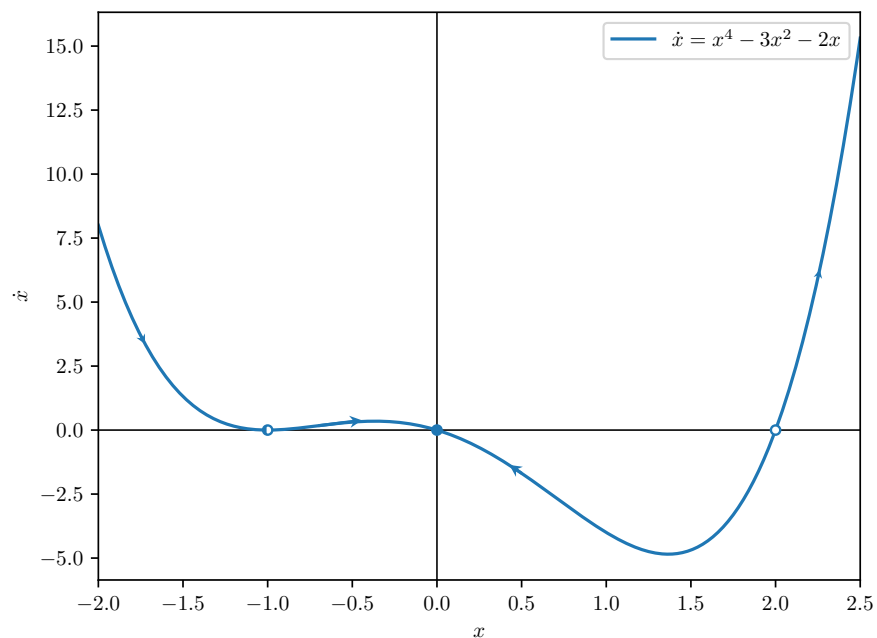


Figure 2.10: Vector field and fixed points for  $\dot{x} = x^4 - 3x^2 - 2x$

Thus, for  $x_1(0) > 1$ , say  $x_1(0) = \frac{3}{2}$ , we have

$$\begin{aligned} \frac{1}{1-c} &= \frac{3}{2} \\ \Leftrightarrow 1-c &= \frac{2}{3} \\ \Leftrightarrow c &= \frac{1}{3}, \end{aligned}$$

so that

$$x_1(t) = \frac{1}{1 - e^t/3}.$$

### 2.2.10

- (a) For the function  $\dot{x} = f(x) = 0$ , every real number is a fixed point, i.e., for  $x \in \mathbb{R}$ ,  $f(x) = 0$ .
- (b) For the function  $\dot{x} = f(x) = \sin \pi x$ , every integer is a fixed point, since  $\forall x \in \mathbb{R}$ ,  $\sin \pi x = 0$ .

- (c) Such a function cannot exist, since  $f(x^*) = 0$  implies that either  $f(x^* - \varepsilon) < 0$  and  $f(x^* + \varepsilon) > 0$  (or vice versa) for some  $\varepsilon \in \mathbb{R}$ ,  $\varepsilon > 0$ , or  $\dot{x} = f(x) = 0 \forall x \in \mathbb{R}$ , by the intermediate value theorem. Thus there cannot be exactly three stable fixed points of a continuous function.
- (d)  $\dot{x} = f(x) = 1$  has no fixed points, since  $\forall x \in \mathbb{R}$ ,  $f(x) \neq 0$ .
- (e) The polynomial

$$\dot{x} = f(x) = (x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_{100}),$$

for  $\alpha_1 \neq \alpha_2, \alpha_1 \neq \alpha_3, \dots, \alpha_2 \neq \alpha_3, \dots, \alpha_{99} \neq \alpha_{100}$  has exactly 100 fixed points.

### 2.2.11

Since we have a linear first-order differential equation, we have the standard form

$$\dot{Q} + \frac{Q}{RC} = \frac{V_0}{R},$$

so that we may identify the integrating factor

$$e^{\int 1/RC dt} = e^{t/RC}$$

and multiply both sides of this equation by the integrating factor to obtain

$$\frac{d}{dt} [e^{t/RC} Q] = e^{t/RC} \frac{V_0}{R}.$$

Integrating both sides yields

$$\begin{aligned} e^{t/RC} Q(t) &= \frac{V_0}{R} \int e^{t/RC} dt \\ &= CV_0 e^{t/RC} + c \\ \Leftrightarrow Q(t) &= CV_0 + ce^{-t/RC}. \end{aligned}$$

Then, using the initial condition  $Q(0) = 0$ , we have

$$c = -CV_0,$$

so that we have the solution

$$Q(t) = CV_0 (1 - e^{-t/RC}).$$